# Supplementary Materials

**Seizure Forecasting and Crash Risk Analysis**

June 3, 2026

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## 1 Overview

This appendix gives the full model, the parameter derivations, the main result, the sensitivity analyses, and the reproducibility instructions for the main paper. All calculations are analytic (closed-form); the accompanying Jupyter notebook `crashes_vs_TiW.ipynb` reproduces every figure and every number in this document.

We quantify the trade-off between seizure-forecasting accuracy (AUC), driving exposure, and the driving restrictions required to maintain fatal crash risk at or below a safety threshold of approximately 6× baseline, corresponding to BAC ≈ 0.05–0.07% for adult drivers. This benchmark is aligned with the 0.05% BAC standard used in Utah and most European countries and is more conservative than the 0.08% BAC limit used in 49 U.S. states.

**Key modeling feature.** Relative fatal crash risk on forecasted "safe" days is computed using a per-hour fatal-crash baseline $\lambda_f \approx 4.5 \times 10^{-7}\,\mathrm{hr}^{-1}$ and a conditional probability $q_f = P(\text{fatal crash} \mid \text{seizure while driving}) = 0.02$ (central estimate; range 0.005–0.05 examined in sensitivity analyses). The probability that a seizure coincides with driving scales as $D/24$ for daily driving duration $D$, but because both the seizure-related risk and the baseline risk scale with exposure, the relative-risk benchmark is approximately invariant to $D$ (Section 2.4).

**Key findings.** Patients with infrequent seizures ($\leq 1$/year) approach unrestricted driving even at low AUC. Patients with frequent seizures require AUC $\geq 0.90$ to enable meaningful driving. At AUC 0.90 and 1 hr/day driving: weekly seizures permit ≈ 53 days/year, monthly seizures ≈ 182, and yearly seizures ≈ 365. These results are largely insensitive to driving duration but moderately sensitive to $q_f$ (Section 4).

# 2 Model and Methods

This section builds the model in the order it is computed: the forecasting model, the seizure prior, the Bayesian posterior on forecast-safe days, the fatal-crash-risk model, the safety threshold, and the conversion to driving days.

## 2.1 Binormal ROC Framework

We model the forecasting algorithm's output as a scalar score $S$ generated by one of two normal distributions, depending on whether the day is a seizure day.

**Variable definitions used throughout this appendix.**

- $S$: the forecasting algorithm's output score on a given day.
- $m$: the separation between the seizure and non-seizure score distributions (larger $m$ means the algorithm discriminates better).
- $t$: the operating threshold; the algorithm warns *unsafe to drive* when $S > t$ and forecasts *safe to drive* when $S \leq t$.
- $\Phi(\cdot)$: the cumulative distribution function (CDF) of the standard normal distribution $N(0,1)$.
- $\Phi^{-1}(\cdot)$: the inverse standard-normal CDF (quantile function).
- AUC: area under the receiver operating characteristic curve.

The forecast score is distributed as

$$S \sim N(0,1) \quad \text{on a non-seizure day,} \qquad S \sim N(m,1) \quad \text{on a seizure day.} \tag{2.1}$$

Under this equal-variance binormal model, the separation $m$ is related to AUC by

$$m = \sqrt{2}\,\Phi^{-1}(\text{AUC}). \tag{2.2}$$

For threshold $t$ applied to the score, the sensitivity (true-positive rate) and false-positive rate are

$$\text{Sensitivity} = P(S > t \mid \text{seizure}) = 1 - \Phi(t - m) = \Phi(m - t), \tag{2.3}$$

$$\text{FPR} = P(S > t \mid \text{no seizure}) = 1 - \Phi(t). \tag{2.4}$$

Figure S1 illustrates the model geometrically: the two score distributions, the threshold $t$, the separation $m$, and the regions whose probability mass equals sensitivity (2.3) and FPR (2.4). Section 4.3 tests robustness to alternative ROC shapes.

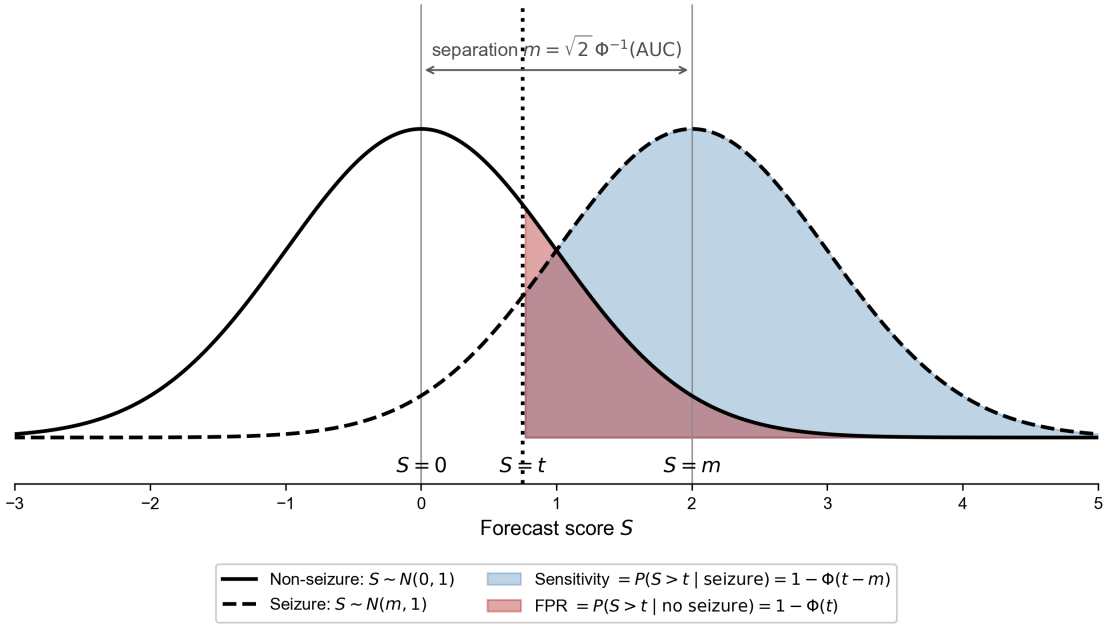



Figure S1: **Binormal forecast model.** Score distributions on non-seizure days ($S \sim N(0, 1)$, solid) and seizure days ($S \sim N(m, 1)$, dashed). The threshold $t$ (dotted vertical) defines the decision boundary; sensitivity (blue) and FPR (red) are the probability masses to the right of $t$ under the seizure and non-seizure distributions, respectively. The separation parameter $m$ relates to AUC via Equation 2.2.

## 2.2 Daily Seizure Probability

For a patient with average seizure rate $R$ (seizures per day), the daily seizure probability under a Poisson process is

$$p = 1 - e^{-R}. \tag{2.5}$$

| Frequency | $R$ | $p$ |
|---|---|---|
| 1 seizure / week | 1/7 | $\approx 0.133$ |
| 1 seizure / month | 1/30 | $\approx 0.033$ |
| 1 seizure / year | 1/365 | $\approx 0.0027$ |

The Poisson mapping is used only to translate an average rate into a daily probability; the analysis is otherwise agnostic to the temporal structure of seizure risk.

### 2.3 Posterior Seizure Probability on Forecast-Safe Days

When a patient drives on a forecast-safe day (one for which the algorithm output $S \leq t$), the residual crash risk is driven by the posterior probability of a seizure given that negative forecast. By Bayes' theorem,

$$P(\text{seizure} \mid \text{forecast safe}) \;=\; \frac{P(\text{forecast safe} \mid \text{seizure}) \cdot p}{P(\text{forecast safe} \mid \text{seizure}) \cdot p + P(\text{forecast safe} \mid \text{no seizure}) \cdot (1-p)}. \tag{2.6}$$

The two conditional probabilities in (2.6) are obtained directly from sensitivity and FPR by complementation:

$$P(\text{forecast safe} \mid \text{seizure}) \;=\; 1 - P(\text{forecast unsafe} \mid \text{seizure}) \;=\; 1 - \text{Sensitivity}, \tag{2.7}$$

$$P(\text{forecast safe} \mid \text{no seizure}) \;=\; 1 - P(\text{forecast unsafe} \mid \text{no seizure}) \;=\; 1 - \text{FPR}. \tag{2.8}$$

Substituting (2.7)–(2.8) into (2.6) and using the binormal expressions for sensitivity (2.3) and FPR (2.4) gives

$$P(\text{seizure} \mid \text{forecast safe}) \;=\; \frac{(1-\text{Sensitivity})\,p}{(1-\text{Sensitivity})\,p + (1-\text{FPR})\,(1-p)} \;=\; \frac{\Phi(t-m)\,p}{\Phi(t-m)\,p + \Phi(t)\,(1-p)}. \tag{2.9}$$

### 2.4 Fatal Crash Risk and the Relative-Risk Derivation

**Parameters used in this subsection.** We use the parameters listed in Table 1 throughout the crash-risk derivation.

Table 1: Crash-risk model parameters.

| Symbol | Meaning | Value | Source |
|---|---|---|---|
| $\lambda_f$ | Baseline fatal-crash rate per hour of sober driving | $4.5 \times 10^{-7}\,\text{hr}^{-1}$ | NHTSA *Passenger Vehicles 2023* + FHWA *Highway Statistics 2023* + AAA *American Driving Survey 2023* (derivation in this section) |
| $q_f$ | $P$(fatal crash \| seizure while driving) | 0.02 central; range 0.005–0.05 | Inferred severity parameter; see Section 4.1 |
| $D$ | Driving hours per day | 0.5, 1, 2 | — |
| $\pi$ | $P$(seizure \| forecast safe) | From Eq. (2.9) | — |

**Baseline per-day fatal-crash risk.** Given $D$ hours of driving per day and a per-hour baseline rate $\lambda_f$, the per-day baseline fatal-crash risk for a sober driver is

$$B(D) \;=\; D \cdot \lambda_f. \tag{2.10}$$

**Derivation of $\lambda_f$.** NHTSA *Passenger Vehicles 2023* reports $\sim 44{,}000$ passenger vehicles involved in fatal crashes; FHWA *Highway Statistics 2023* Table VM-1 reports $\sim 2.88$ trillion light-duty vehicle-miles travelled (VMT). The ratio is approximately 1.5 passenger-vehicle fatal-crash involvements per $10^8$ miles, i.e. $1.5 \times 10^{-8}$ per mile. Converting per-mile to per-hour requires a driving speed: the AAA *American Driving Survey 2023* reports $\approx 29.1$ miles over $\approx 60.7$ minutes per driver per day, i.e. $\approx 29$ mph. Then

$$\lambda_f \;\approx\; 1.5 \times 10^{-8}\,\text{mile}^{-1} \times 29\,\text{mph} \;\approx\; 4.4 \times 10^{-7}\,\text{hr}^{-1} \;\approx\; 4.5 \times 10^{-7}\,\text{hr}^{-1}. \tag{2.11}$$

(FHWA Table VM-1 supplies the VMT denominator only; the driving speed used for the per-mile-to-per-hour conversion comes from the AAA survey.) Because this baseline incorporates all non-seizure causes of fatal crashes (distraction, weather, other drivers), the model conservatively attributes this background risk to every driving day regardless of seizure status.

**Seizure-day fatal-crash risk.** On a day when a seizure occurs, the per-day fatal-crash risk depends on whether the seizure happens to fall within the driving window. Assuming uniform seizure timing across the 24-hour day,

$$P(\text{fatal crash} \mid \text{seizure day}) \;=\; \frac{D}{24} \cdot q_f \;+\; \left(1 - \frac{D}{24}\right) \cdot B(D). \tag{2.12}$$

The first term is the contribution when the seizure happens while driving; the second is when the seizure happens outside driving hours and the day's driving carries only baseline risk.

**Forecast-safe-day fatal-crash risk.** Let $\pi = P(\text{seizure} \mid \text{forecast safe})$ from Equation 2.9. The residual fatal-crash risk on a forecast-safe day is

$$P(\text{fatal crash} \mid \text{forecast safe}) \;=\; \pi \cdot P(\text{fatal crash} \mid \text{seizure day}) \;+\; (1-\pi) \cdot B(D). \tag{2.13}$$

**Relative fatal-crash risk and the exposure-invariance derivation.** Substituting (2.12) into (2.13) and dividing by the baseline $B(D)$ defines the relative risk:

$$\text{RR}(D) \;=\; \frac{P(\text{fatal crash} \mid \text{forecast safe})}{B(D)} \;=\; \frac{\pi\left[\dfrac{D}{24}\, q_f + \left(1 - \dfrac{D}{24}\right) B(D)\right] + (1-\pi)\, B(D)}{B(D)}. \tag{2.14}$$

Distribute the numerator and divide each term by $B(D) = D\lambda_f$:

$$\text{RR}(D) \;=\; \pi \cdot \frac{(D/24)\, q_f}{D\, \lambda_f} \;+\; \pi\left(1 - \frac{D}{24}\right) \;+\; (1-\pi). \tag{2.15}$$

The $D$ in the first term cancels:

$$\text{RR}(D) \;=\; \pi \cdot \frac{q_f}{24\, \lambda_f} \;+\; \pi \;-\; \pi \cdot \frac{D}{24} \;+\; 1 - \pi. \tag{2.16}$$

The $\pi$ and $-\pi$ cancel, leaving the exact expression

$$\boxed{\mathrm{RR}(D) \;=\; 1 + \pi\left[\frac{q_f}{24\,\lambda_f} - \frac{D}{24}\right]}. \tag{2.17}$$

Under our parameters, $q_f/(24\lambda_f) = 0.02/(24 \times 4.5 \times 10^{-7}) \approx 1852$, while $D/24 \le 2/24 \approx 0.083$ for $D \le 2$ hours. Since $1852 \gg 0.083$, the $D/24$ term is negligible and we approximate

$$\mathrm{RR}(D) \;\approx\; 1 + \pi \cdot \frac{q_f}{24\,\lambda_f} \;=\; 1 + \pi \cdot A, \qquad A \;\equiv\; \frac{q_f}{24\,\lambda_f}. \tag{2.18}$$

$A$ is the *amplification factor*.

**Exposure invariance.** The approximation (2.18) makes the $D$-dependence vanish: at $D = 1$ hr, the residual $D/24$ correction is at most $0.083\,\pi$ versus the dominant $\sim 1852\,\pi$ (a relative discrepancy of order $5 \times 10^{-5}$). Absolute fatal-crash risk does scale with driving exposure through $B(D)$, but the *relative* risk against the BAC-anchored threshold is approximately invariant to $D$. Section 4.2 confirms this numerically.

## 2.5 Safety Threshold

We use a safety threshold derived from the BAC 0.05% standard. Under the doubling-per-0.02-BAC parametric model of Zador et al. and Compton & Berning, the relative fatal-crash risk at BAC 0.05% is

$$T \;=\; 2^{\,0.05/0.02} \;\approx\; 5.66, \tag{2.19}$$

which is approximately 6× baseline. The prose throughout this appendix describes this as a "6× threshold" for readability; the code uses the precise value $T = 2^{0.05/0.02}$. We find the operating threshold $t$ such that

$$\frac{P(\text{fatal crash} \mid \text{forecast safe})}{B(D)} \;\le\; T, \tag{2.20}$$

and report the corresponding minimum days in warning (the minimum restriction required to meet the threshold).

## 2.6 Time in Warning and Driving Days

The proportion of days the algorithm flags as high-risk (the "time in warning") is

$$P(\text{warning}) \;=\; \text{Sensitivity} \cdot p + \text{FPR} \cdot (1 - p), \tag{2.21}$$

$$\text{Days in warning per year} \;=\; P(\text{warning}) \times 365. \tag{2.22}$$

The complement, $(1 - P(\text{warning})) \times 365$, is the number of days per year on which the patient is forecast safe to drive. An algorithm that warns every day achieves perfect sensitivity but permits zero driving days; the analysis finds, for each AUC, the operating threshold $t$ that just meets the safety constraint (2.20) and reports the resulting driving days.

Table 2: Driving restrictions required to achieve the safety threshold by seizure frequency and forecasting performance (1 hr/day driving).

| Seizure freq. | AUC | Warning days/yr | Driving days/yr | Avg. days between drives |
|---|---|---|---|---|
| **1/week** | 0.60 | 365 | 0 | Cannot reach safety threshold |
| | 0.70 | 365 | 0 | Cannot reach safety threshold |
| | 0.80 | 363 | 2 | $\sim 237$ |
| | 0.90 | 312 | 53 | 6.9 |
| | 0.95 | 221 | 144 | 2.5 |
| | 0.99 | 91 | 274 | 1.3 |
| **1/month** | 0.60 | 365 | 0 | Cannot reach safety threshold |
| | 0.70 | 364 | 1 | $\sim 497$ |
| | 0.80 | 323 | 42 | 8.8 |
| | 0.90 | 183 | 182 | 2.0 |
| | 0.95 | 92 | 273 | 1.3 |
| | 0.99 | 23 | 342 | 1.1 |
| **1/year** | 0.60 | 34 | 331 | 1.1 |
| | 0.70 | 8 | 357 | 1.0 |
| | 0.80 | 2 | 363 | 1.0 |
| | 0.90 | 0 | 365 | 1.0 (already safe) |
| | 0.95 | 0 | 365 | 1.0 (already safe) |
| | 0.99 | 0 | 365 | 1.0 (already safe) |

# 3 Main Result

Table 2 reports the driving restrictions required to achieve the safety threshold by seizure frequency and forecasting performance, at 1 hr/day driving, $q_f = 0.02$, $\lambda_f = 4.5 \times 10^{-7}\,\text{hr}^{-1}$.

Figure S2 shows the underlying mechanism: how crash risk on forecast-safe days falls as the algorithm spends more days in warning, and where each AUC curve crosses the safety threshold.

**Key insights.**

- *High AUC is the binding constraint.* Algorithms need AUC $\geq 0.90$ to provide meaningful driving days for patients with weekly or monthly seizures.
- *Patients with rare seizures ($\leq$ 1/year)* are at or near unrestricted driving even at low AUC ($\approx$ 331 days/year at AUC 0.60), though not literally always-safe.
- *Monthly seizure patients benefit greatly*: AUC 0.90 permits $\approx$ 182 days/year; AUC 0.95 permits $\approx$ 273.
- *Weekly seizure patients face significant restrictions*: even AUC 0.90 permits only $\approx$ 53 days/year; AUC 0.80 is essentially insufficient ($\leq$ 2 days/year).

# 4 Sensitivity Analyses

Three analyses test the robustness of the main result: to the most uncertain parameter ($q_f$, Section 4.1), to driving duration (Section 4.2), and to the shape of the ROC curve (Section 4.3).

### 4.1 Conditional Fatal-Crash Probability $q_f$

The parameter $q_f = P(\text{fatal crash} \mid \text{seizure while driving})$ is the most uncertain component of the model and is not directly reported by any single study; it is an inferred severity parameter. We examined the defensible range $q_f \in [0.005,\ 0.05]$:

- **Lower bound ($q_f = 0.005$):** moderate crash rate ($\sim 0.30$) given a seizure while driving $\times$ moderate case-fatality ($\sim 0.02$) — closer to baseline fatal-crash severity.
- **Central ($q_f = 0.02$):** Gastaut & Zifkin 1987 observed crash rate ($\sim 0.55$) $\times$ case-fatality of seizure-related crashes ($\sim$ 3–5%, inferred from Sheth et al. 2004 and Drazkowski et al. 2003).
- **Upper bound ($q_f = 0.05$):** counts near-misses as effective crashes (Gastaut: $\sim 38\%$ additional near-miss rate) with upper-bound severity.

Table 3 reports driving days/year at AUC = 0.90, 1 hr/day, across the $q_f$ range.

Table 3: Sensitivity of driving days/year to $q_f$ at AUC = 0.90, 1 hr/day.

| Seizure frequency | $q_f = 0.005$ | $q_f = 0.02$ (central) | $q_f = 0.05$ |
|---|---|---|---|
| 1/week | 154 | 53 | 20 |
| 1/month | 314 | 182 | 99 |
| 1/year | 365 | 365 | 334 |

The amplification factor $A = q_f/(24\lambda_f)$ ranges from $\approx 463$ (lower) through $\approx 1852$ (central) to $\approx 4630$ (upper); the corresponding maximum tolerable $P(\text{seizure} \mid \text{forecast safe})$ for RR $\leq T = 5.66$ ranges from $\approx 1.0\%$ down to $\approx 0.10\%$.

**Interpretation.** Across the full range, the qualitative finding is preserved: weekly-seizure patients cannot achieve unrestricted driving even at AUC 0.90, and yearly-seizure patients remain at or near unrestricted driving. The quantitative allowance for monthly seizures spans roughly a factor of three. Stakeholders may prefer the conservative-end $q_f = 0.05$ until the parameter is better empirically anchored.

### 4.2 Driving Duration (Exposure Invariance)

Section 2.4 shows analytically that relative risk is approximately invariant to driving duration $D$, because both the seizure-related and baseline fatal risks scale with exposure (Equation 2.18). Numerically, the driving-day allowances coincide to integer rounding across the examined durations (Table 4).

Table 4: Driving days/year coincide across exposure levels under the relative-risk benchmark.

| Seizure frequency | AUC | 30 min/day | 1 hr/day | 2 hr/day |
|---|---|---|---|---|
| 1/week | 0.90 | 53 | 53 | 53 |
| 1/month | 0.90 | 182 | 182 | 182 |
| 1/year | 0.80 | 363 | 363 | 363 |

Absolute fatal-crash risk does still scale with driving duration; an alternative *absolute*-risk benchmark would depend directly on exposure. Under the relative-risk benchmark used here, driving duration is not a material determinant of the allowable driving days.

### 4.3 ROC Curve Shape

The equal-variance binormal assumption implies a specific sensitivity–specificity relationship. Real ROC curves can deviate due to unequal variances (S-shaped), mixture distributions (hooked; some seizures easier to predict than others), or finite-sample effects. We test three score-generating models, all calibrated to AUC $\approx 0.90$.

**Equal-variance binormal** (baseline): negatives $N(0,1)$, positives $N(m,1)$, with $m = \sqrt{2}\,\Phi^{-1}(0.90) \approx 1.81$. Sensitivity and FPR are given by Eqs. (2.3)–(2.4).

**Unequal-variance binormal** (S-shaped): negatives $N(0,1)$, positives $N(\mu_1, \sigma_1)$ with $\mu_1 = 2.4$, $\sigma_1 = 1.4$. For threshold $t$,

$$\text{Sensitivity} \;=\; 1 - \Phi\left(\frac{t-\mu_1}{\sigma_1}\right), \qquad \text{FPR} \;=\; 1 - \Phi(t). \tag{4.1}$$

The posterior on forecast-safe days uses $\Phi((t-\mu_1)/\sigma_1)$ in place of $\Phi(t-m)$ in Equation 2.9.

**Mixture model** (hooked): negatives $N(0,1)$, positives $w\,N(\mu_{\text{easy}},1) + (1-w)\,N(\mu_{\text{hard}},1)$, with $w = 0.7$, $\mu_{\text{easy}} = 2.5$, $\mu_{\text{hard}} = 0.8$. Then

$$\text{Sensitivity} \;=\; w\,[1-\Phi(t-\mu_{\text{easy}})] \;+\; (1-w)\,[1-\Phi(t-\mu_{\text{hard}})], \tag{4.2}$$

$$P(S < t \mid \text{seizure}) \;=\; w\,\Phi(t-\mu_{\text{easy}}) \;+\; (1-w)\,\Phi(t-\mu_{\text{hard}}). \tag{4.3}$$

Table 5 reports driving days at AUC $\approx 0.90$ and 1 hr/day for each ROC shape. Figure S5 shows the crash-risk-versus-warning-days curves for each shape and seizure frequency.

Table 5: Driving days/year at AUC $\approx 0.90$, 1 hr/day, by ROC shape.

| Seizure frequency | Equal-variance binormal | S-shaped (unequal var.) | Hooked (mixture) |
|---|---|---|---|
| 1/week | 53 | Cannot reach safety | 1 |
| 1/month | 182 | 145 | 66 |
| 1/year | 365 | 365 | 365 |

**Findings.**

- *At high seizure frequencies, ROC shape matters.* Equal-variance binormal is the most optimistic shape; the S-shaped form fails to reach the threshold at all for weekly seizures, and the hooked mixture model permits only 1 day. For monthly seizures, the spread is 182 → 145 → 66 days/year.

- *At low seizure frequency (yearly), shape is irrelevant* — all three ROC families permit essentially full-year driving because the prior seizure probability is already very low.
- *Net.* AUC remains the primary determinant, but ROC geometry is a meaningful secondary factor. Because equal-variance binormal is the most optimistic of the three, real-world (S-shaped or hooked) ROCs at the same AUC will generally permit *fewer* driving days than the main result suggests — so reporting AUC alone may overstate clinical utility for patients with frequent seizures. The qualitative conclusion (high AUC is necessary, and even AUC 0.90 is borderline for weekly seizures) is robust to shape.

# 5 Assumptions, Limitations, and Recommendations

**Why the binormal model is a reasonable base case.** It is a standard approximation in diagnostic testing and signal-detection theory, is mathematically tractable (closed-form, reproducible), and for a given AUC tends to offer near-best-case sensitivity-specificity trade-offs — giving the forecasting algorithm the benefit of the doubt. The paper's question is "what level of performance (AUC) is needed for safe driving decisions?", not "how does one specific empirical ROC curve perform?". Section 4.3 quantifies the effect of departures from this assumption.

**Driving-exposure assumptions.**

1. **Seizure timing is uniformly distributed** over the 24-hour day. Many patients have circadian patterns (nocturnal epilepsy, morning clustering); the model is conservative for seizures concentrated outside driving hours and optimistic for daytime clustering.
2. **Driving duration is constant across days.** Real driving varies; the 30 min/day–2 hr/day range brackets typical commuter and heavy-driver patterns (and, per Section 4.2, barely affects the relative-risk result).
3. $q_f = 0.02$ **is an inferred severity parameter** (range 0.005–0.05), decomposing as $P(\text{crash} \mid \text{seizure while driving}) \approx 0.55$ (Gastaut & Zifkin 1987) $\times$ case-fatality $\approx 3$–$5\%$ (inferred from Sheth et al. 2004 and Drazkowski et al. 2003). It depends on seizure type, aura, road conditions, and vehicle safety features; Section 4.1 reports the sensitivity analysis.

**Recommendations for algorithm developers.**

1. Aim for AUC $\geq 0.90$ as a minimum for driving applications.
2. Characterise your empirical ROC shape — an S-shaped or hooked curve at the same AUC permits fewer driving days and shifts optimal operating points.
3. For patients with frequent seizures, even AUC $= 0.90$ may be insufficient for practical driving policies.
4. For patients with rare seizures, forecasting may be unnecessary for safety; focus on other quality-of-life benefits.
5. Consider patient-specific seizure frequency and driving patterns when counselling about risk.

# 6 Reproducibility

All calculations use NumPy, SciPy (`scipy.stats.norm` for the Gaussian CDF/inverse CDF), and Matplotlib, with analytic closed-form expressions.

```
pip install numpy matplotlib scipy
jupyter notebook crashes_vs_TiW.ipynb        # interactive
# or run all cells programmatically:
jupyter nbconvert --to notebook --execute crashes_vs_TiW.ipynb
```

Table 6: Notebook cell layout and outputs.

| Cell | Produces |
|---|---|
| 0 | Figure S2 — crash risk vs. days in warning (3 frequencies, 1 hr/day) |
| 1 | Figure 1 — driving days vs. AUC (main result) |
| 2 | Figure S1 — binormal model schematic |
| 3 | Figures S4 & S5 — ROC shape geometries and sensitivity |
| 4–5 | Extended AUC table (tab-delimited, all frequencies/durations) |
| 6 | Figure S3 — $q_f$ sensitivity |

Figure files are produced as PDF and PNG: `Figure_1`, `Figure_S1`–`Figure_S5`.

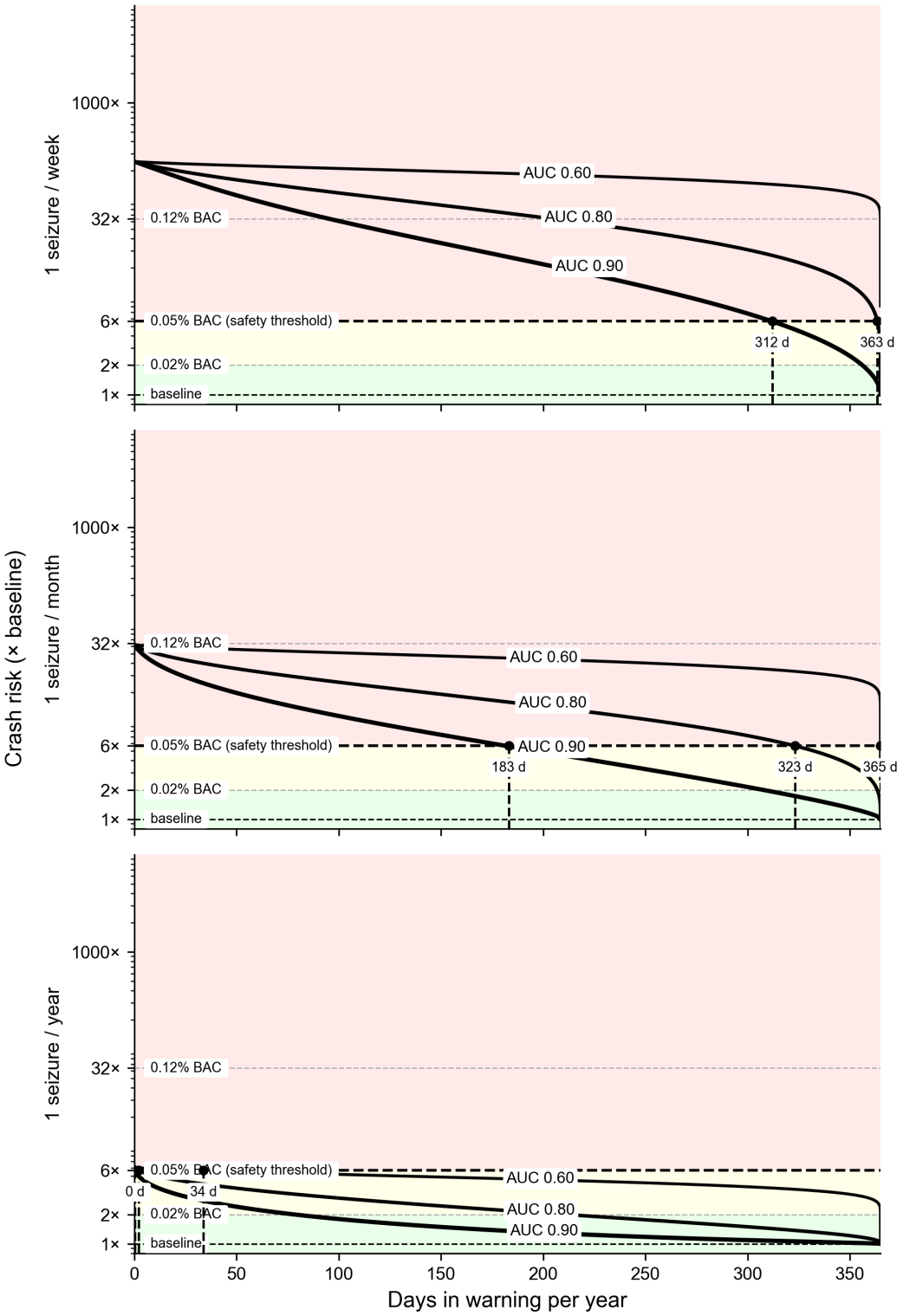



Figure S2: **Mechanism: crash risk versus days in warning.** Rows are seizure frequencies (1/week, 1/month, 1/year), at 1 hr/day driving. Black curves are AUC 0.60, 0.80, 0.90 (thicker lines indicate higher AUC). Shaded bands and dashed lines mark BAC-equivalent risk levels (0.02%, the 0.05% safety threshold, 0.12%). Black circles mark the minimum days in warning needed to reach the threshold.

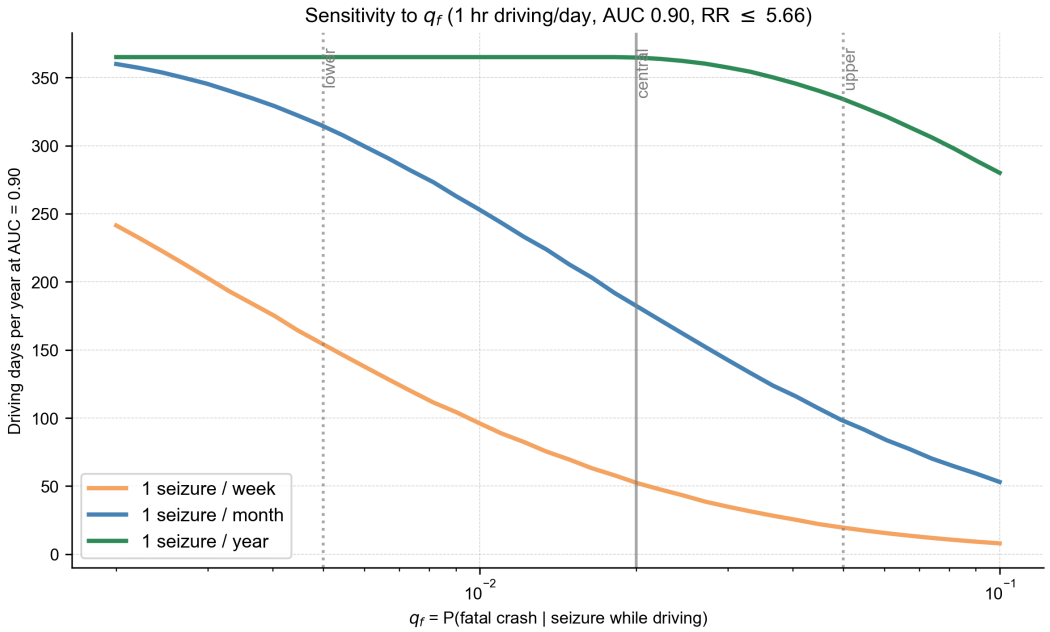



Figure S3: **Sensitivity of driving days to $q_f$.** Driving days per year at AUC = 0.90 and 1 hr/day, as a function of $q_f$. Vertical guides mark the lower bound (0.005), the central estimate (0.02), and the upper bound (0.05).

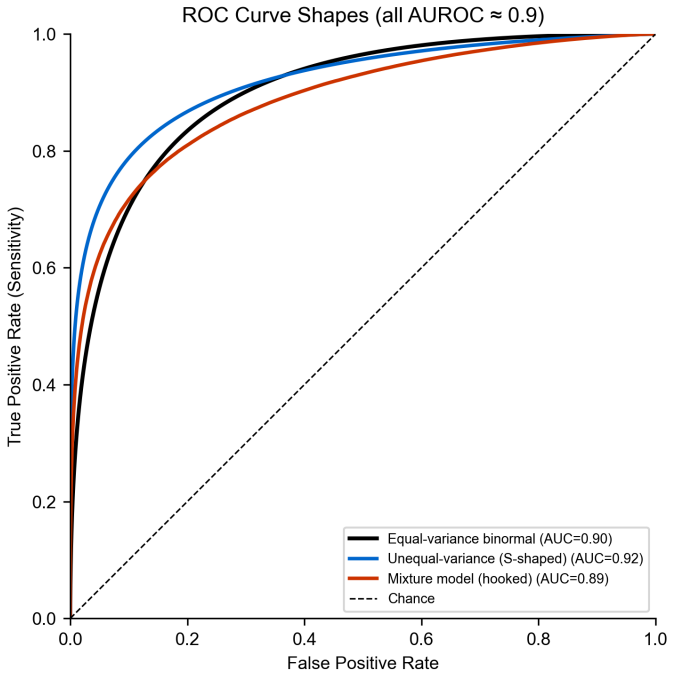



Figure S4: **Three ROC curve shapes**, all at AUROC ≈ 0.90. Equal-variance binormal (black, central), unequal-variance/S-shaped (blue, top-left), and hooked mixture model (red).

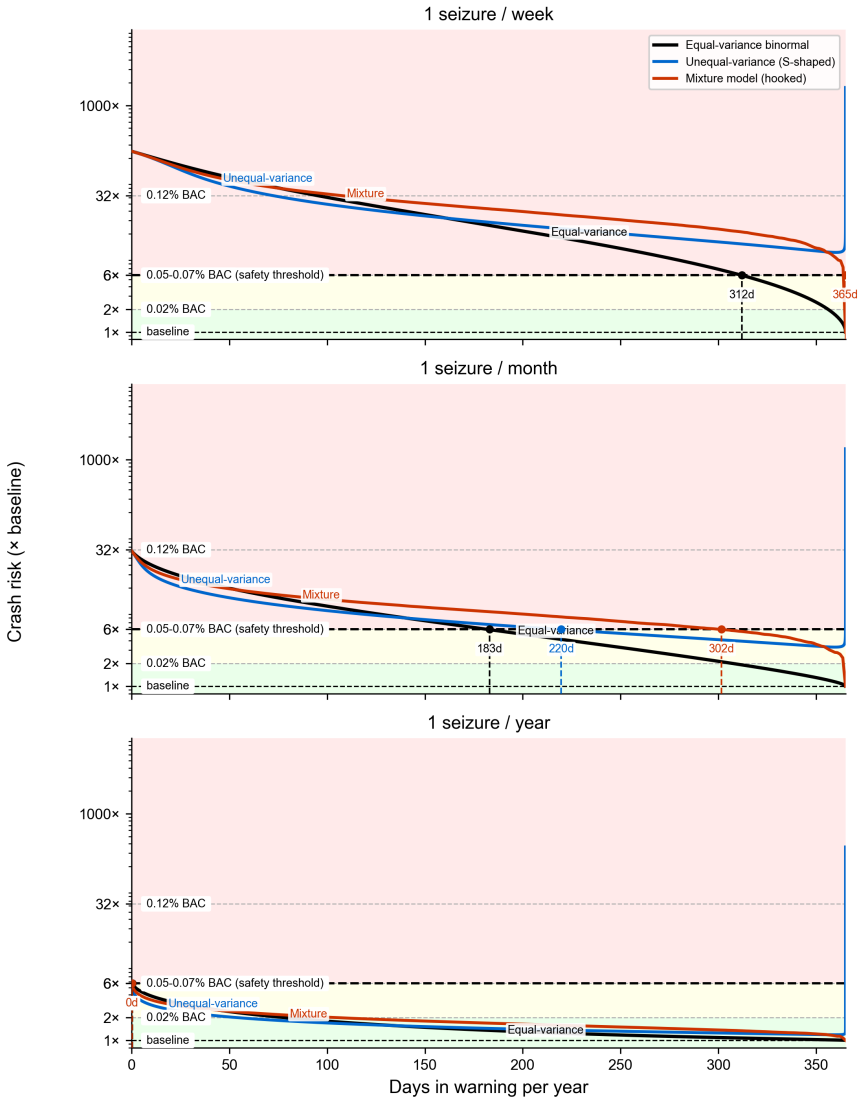



Figure S5: **Crash risk versus days in warning for three ROC shapes**, by seizure frequency, at 1 hr/day and AUC $\approx 0.90$. Equal-variance binormal (black), S-shaped (blue), and hooked mixture (red). Black dashed line: safety threshold.